Stability of QAOA Warm-Starts under Graph Updates

Lemmas A, B, Theorem C, Proposition D and Lemma D0 — draft for supervisor review

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Status of this document. Draft proofs for the theory chapter of the bachelor thesis *Learned and Cached Warm-Starts for QAOA on Weighted Max-Cut with an Application to Dynamic Vehicle Routing*. The proofs were drafted with AI assistance (Claude) and are being checked by the author; every statement was also tested numerically (summary below). Comments and corrections are welcome.

Overview

In the dynamic vehicle-routing setting, each new customer order adds one vertex to a complete, weighted clustering graph, and QAOA (depth p) is re-optimised. The question is how much the objective and its optimal angles move, which decides whether the previous angles are a good starting point.

Result	Regime	Statement (informal)	Numerical check
Lemma A0	any	adding an isolated vertex does not change F	exact equality
Lemma A, Cor. A1	dense (complete graph)	$\lvert\Delta F\rvert \le d_x\,(1 + (C_{\max} - C_{\min}) \sum_l \lvert\gamma_l\rvert)$	always held; actual/bound median 0.07, max 0.49
Lemma B, Cor. B1	sparse	each edge term depends only on its distance-$(p-1)$ light cone; a vertex at distance $\ge p+1$ changes nothing	exact; threshold sharp
Cor. B2	sparse, degree $\le D$	$\lvert\Delta F\rvert \le \sum_{e \in A} w_e$, $\lvert A\rvert = O(D^{p+1})$, independent of n	—
Theorem C	any, curvature $\mu > 0$	$\lVert \theta'^* - \theta^* \rVert \le \varepsilon/\mu$	held in 38/38 valid cases; ratio median 0.31, max 0.68
Proposition D	any, curvature $\mu > 0$	one Newton step: error $\le \frac{M}{2\mu} \lVert \theta'^* - \theta^* \rVert^2$	error/shift^2 bounded (median 2.4, max 3.7); real orders: error 0.08 of reuse
Lemma D0	new vertex, any graph	$F_t - F_{-t} = tW_v$, so the angle shift is $O(t^2)$ in the new edge weights	exact identity; log-log slope 1.97

Open points. (i) Explicit constants for ε in Theorem C are only sketched. (ii) Lemma A is loose (about 14 times the actual change); tightening it is an open question. (iii) Novelty: the light cone is known (Farhi et al. 2014) and Lemma A / Theorem C use standard perturbation arguments; the intended contribution is their combination for dynamic QAOA and the measured comparison of the two regimes.

Modelling choice used throughout. The routing graph is complete with distance weights on a fixed scale for the whole stream, so that maximising the cut equals minimising the total within-group distance (cut + within = W). A k-nearest-neighbour graph was rejected: Max-Cut on it separates neighbouring customers (checked numerically).

Lemma A: stability of the QAOA objective when a vertex is added (dense regime)

Setting and notation

Graph $G = (V, E, w)$ with $|V| = n$. Cost and mixer Hamiltonians:

$$C = \sum_{(u,v) \in E} w_{uv} \frac{I - Z_u Z_v}{2}, \qquad B = \sum_{j \in V} X_j .$$

For $\theta = (\gamma_1, \ldots, \gamma_p, \beta_1, \ldots, \beta_p)$:

$$U_C(\theta) = V_p \cdots V_1, \quad V_l = e^{-i\beta_l B} e^{-i\gamma_l C}, \qquad |\psi_C(\theta)\rangle = U_C(\theta) |+\rangle^{\otimes n}, \qquad F_C(\theta) = \langle \psi_C(\theta) | C | \psi_C(\theta) \rangle .$$

$\| \cdot \|$ is the Euclidean norm on vectors and the operator norm on matrices. For a diagonal matrix D, $\| D \| = \max_z |D_{zz}|$. Write $C_{\max}, C_{\min}$ for the largest and smallest cut values.

Lemma A0 (an isolated vertex changes nothing)

Let G^+ be G plus a vertex x with no edges. Then $F_{C^+}(\theta) = F_C(\theta)$ for every θ.

Proof. $C^+ = C \otimes I_x$ and $B^+ = B \otimes I_x + I \otimes X_x$. The two terms of B^+ commute, so $e^{-i\beta B^+} = e^{-i\beta B} \otimes e^{-i\beta X_x}$, and $e^{-i\gamma C^+} = e^{-i\gamma C} \otimes I_x$. Hence $|\psi_{C^+}\rangle = |\psi_C\rangle \otimes |\phi\rangle$ with $|\phi\rangle = \prod_l e^{-i\beta_l X_x} |+\rangle$, a unit vector. Then $F_{C^+} = \langle \psi_C | C | \psi_C \rangle \langle \phi | \phi \rangle = F_C$. ▫

Lemma A (perturbation of the cost Hamiltonian)

Let C and $C' = C + \Delta C$ be cost Hamiltonians on the same n qubits (both diagonal in the computational basis). Then for every θ and every real constant c,

$$|F_{C'}(\theta) - F_C(\theta)| \le \| \Delta C \| \left(1 + 2 \| C - cI \| \sum_{l=1}^{p} |\gamma_l|\right).$$

With $c = (C_{\max} + C_{\min})/2$ this gives

$$|F_{C'}(\theta) - F_C(\theta)| \le \| \Delta C \| \left(1 + (C_{\max} - C_{\min}) \sum_{l=1}^{p} |\gamma_l|\right).$$

Proof. Write $\psi = \psi_C(\theta)$, $\psi' = \psi_{C'}(\theta)$.

Step 1 (split).

$$F_{C'} - F_C = \langle \psi' | \Delta C | \psi' \rangle + (\langle \psi' | C | \psi' \rangle - \langle \psi | C | \psi \rangle).$$

Since $\| \psi' \| = 1$, $|\langle \psi' | \Delta C | \psi' \rangle| \le \| \Delta C \|$.

Step 2 (difference of expectations). For unit vectors a, b and Hermitian A,

$$\langle a|A|a\rangle - \langle b|A|b\rangle = \langle a-b|A|a\rangle + \langle b|A|a-b\rangle$$

(expand the right side: the cross terms $\pm\langle b|A|a\rangle$ cancel). By Cauchy–Schwarz, $|\langle a|A|a\rangle - \langle b|A|b\rangle| \le 2 \| A \| \, \| a-b \|$. Because $\langle a|a\rangle = \langle b|b\rangle = 1$, replacing A by $A - cI$ does not change the left side, so we may use $\| C - cI \|$ in place of $\| C \|$. Hence the second term of Step 1 is at most $2 \| C - cI \| \, \| \psi' - \psi \|$.

Step 3 (distance between the two states). $\| \psi' - \psi \| \le \| U_{C'} - U_C \|$ because $|+\rangle^{\otimes n}$ is a unit vector. For products of unitaries, $\| V'_p \cdots V'_1 - V_p \cdots V_1 \| \le \sum_l \| V'_l - V_l \|$; by induction,

$$U'_k - U_k = V'_k \, (U'_{k-1} - U_{k-1}) + (V'_k - V_k) \, U_{k-1},$$

and unitaries have norm 1. The mixer factor is the same in V_l and V'_l, so

$$\| V'_l - V_l \| = \| e^{-i\gamma_l C'} - e^{-i\gamma_l C} \| = \| e^{-i\gamma_l C} \, (e^{-i\gamma_l \Delta C} - I) \| = \max_z |e^{-i\gamma_l \, \Delta C_{zz}} - 1| \le |\gamma_l| \, \| \Delta C \|,$$

using that C and ΔC are diagonal (so they commute and $e^{-i\gamma(C+\Delta C)} = e^{-i\gamma C} \, e^{-i\gamma \Delta C}$) and $|e^{it} - 1| \le |t|$. Summing, $\| \psi' - \psi \| \le \| \Delta C \| \sum_l |\gamma_l|$.

Step 4 (combine). Steps 1–3 give the first inequality. For the second, $C - cI$ is diagonal with entries $C_{zz} - c \in [C_{\min} - c, \, C_{\max} - c]$, so $\| C - cI \| = (C_{\max} - C_{\min})/2$ for $c = (C_{\max} + C_{\min})/2$. ▫

Corollary A1 (new order in the dense DVRP graph)

Let G have n vertices and let G' add a vertex x joined to every old vertex with weights $w_{xi} \ge 0$, keeping the old weights unchanged. Then for every θ

$$|F_{G'}(\theta) - F_G(\theta)| \le d_x \, (1 + (C_{\max} - C_{\min}) \sum_l |\gamma_l|), \qquad d_x = \sum_i w_{xi} \le n \max_i w_{xi}.$$

Proof. By Lemma A0, compare G' with G^+ (same F as G). On $n+1$ qubits, $\Delta C = \sum_i w_{xi} (I - Z_x Z_i)/2$ is diagonal with entries $\sum_i w_{xi} [z_x \ne z_i] \in [0, d_x]$, so $\| \Delta C \| \le d_x$. Apply Lemma A ($C_{\max}, C_{\min}$ of G^+ equal those of G). ▫

Remarks

1. **How tight is it?** On 200 random complete Euclidean graphs ($n = 8 \to 9$, $p = 1$, $\gamma \in [0, 1]$), the ratio (actual change)/(bound) had median 0.07 and maximum 0.49 (experiment E1). So the bound is valid but typically about 14 times larger than the actual change. Tightening it is open.
2. **Relative size.** F itself is of order $W = \sum w$, which grows like n^2 on a complete graph, while d_x grows like n. The first term is therefore $O(1/n)$ relative to F. The second term is small only if $(C_{\max} - C_{\min}) \sum |\gamma_l|$ is bounded; whether good angles scale like $1/W$ is **a conjecture to test** (compare Sureshbabu et al. 2024 on weight rescaling).
3. **Weight normalisation.** If weights are re-normalised by a new maximum distance, every old weight is scaled by $s \le 1$ and $\Delta C = (s-1)C + \Delta C_{\text{new}}$, so $\| \Delta C \| \le (1-s) \| C \| + d_x$. To keep Corollary A1

clean, fix the scale for the whole stream (decision D8).

4. **Novelty.** The argument is a standard perturbation bound; do not claim it as new without a literature check. What can be new is its use for dynamic DVRP streams and the comparison with the sparse (light-cone) regime.
5. **From F to the optimal angles.** Lemma A bounds the change of the objective, not of θ^*. Theorem C (next) adds a curvature assumption at θ^* to bound $\| \theta^*(G') - \theta^*(G) \|$.

Lemma B: locality (light cone) of the QAOA objective (sparse regime)

Notation

As in Lemma A. For an edge $e = (u,v)$ write $C_e = w_e (I - Z_u Z_v)/2$, so $F_C = \sum_e \langle C_e \rangle$. dist (x, S) is the graph distance from vertex x to the set S. The *support* supp (O) of an operator is the set of qubits it acts on non-trivially. $N[S] = S \cup \{y: y \text{ adjacent to } S\}$ is the closed neighbourhood.

Lemma B

Fix an edge $e = (u,v)$ and depth p. Let $E_e^{(p)}$ be the set of edges with at least one endpoint at distance $\le p - 1$ from $\{u,v\}$. Then $\langle \psi(\theta) | C_e | \psi(\theta) \rangle$ depends only on the weights of the edges in $E_e^{(p)}$: replacing G by the graph with edge set $E_e^{(p)}$ (same vertices) does not change it, for every θ.

Proof. Heisenberg picture: $\langle C_e \rangle = \langle + | O_0 | + \rangle$ with $O_p = C_e$ and $O_{l-1} = V_l^\dagger O_l V_l$, $V_l = e^{-i\beta_l B} e^{-i\gamma_l C}$, for $l = p, \ldots, 1$.

Claim: supp $(O_{l-1}) \subseteq N[S_l]$ where $S_l =$ supp (O_l), and O_{l-1} depends only on weights of edges touching S_l.

a. Mixer. $e^{i\beta B} = \bigotimes_j e^{i\beta X_j}$ is a product of one-qubit unitaries. Factors on qubits outside S_l commute with O_l and cancel with their inverses, so $e^{i\beta B} O_l e^{-i\beta B}$ is still supported on S_l.

b. Phase. Split $C = C_{\text{in}} + C_{\text{out}}$, where C_{in} collects the edges with at least one endpoint in S_l and C_{out} the rest. C_{out} acts only outside S_l, so it commutes with the (mixed) operator; it also commutes with C_{in} (all terms are diagonal). Hence $e^{i\gamma C} P e^{-i\gamma C} = e^{i\gamma C_{\text{in}}} P e^{-i\gamma C_{\text{in}}}$, which is supported on supp $(C_{\text{in}}) \cup S_l \subseteq N[S_l]$ and involves only edges touching S_l.

(Order inside $V_l^\dagger O V_l = e^{i\gamma C} e^{i\beta B} O e^{-i\beta B} e^{-i\gamma C}$: first (a), then (b).)

Starting from $S_p = \{u,v\}$ and applying the claim p times, the layer l step uses only edges touching $S_l \subseteq \{y:$ dist $(y, \{u,v\}) \le p - l\}$. Over all layers these are edges with an endpoint at distance $\le p - 1$, i.e. $E_e^{(p)}$. Finally $|+\rangle^{\otimes n}$ is a product state, so $\langle + | O_0 | + \rangle$ depends only on O_0. ▫

Corollary B1 (a new order far away changes nothing)

Add a vertex x with any edges. If $\mathrm{dist}_{G'}(x, \{u,v\}) \geq p+1$, then $\langle C_e \rangle$ is unchanged for every θ.

Proof. A new edge (x,i) is in $E_e^{(p)}$ only if x or i is within $p-1$ of $\{u,v\}$, which forces $\mathrm{dist}(x,\{u,v\}) \leq p$. Distances among vertices within $p-1$ of $\{u,v\}$ are not shortened by x (a path through x has length $\geq p+1$ to reach x). So $E_e^{(p)}$ is the same in G and G'; by Lemma A0 the extra qubit does not matter. ▫ The threshold is sharp: at distance exactly p the term changes in general (tested on a path graph).

Corollary B2 (bound with bounded degree)

Let all weights be non-negative and let A be the old edges with $\mathrm{dist}(x,e) \leq p$, plus the new edges. Then

$$|F_{G'}(\theta) - F_G(\theta)| \leq \sum_{e \in A} w_e .$$

If every vertex has degree $\leq D$, then $|A| \leq D\,|B_p(x)| \leq D(1 + D\sum_{k=0}^{p-1}(D-1)^k) = O(D^{p+1})$, **independent of** n.

Proof. Terms outside A are unchanged (B1). Each $\langle C_e \rangle \in [0, w_e]$ because $0 \leq C_e \leq w_e I$, so each changed (or new) term moves by at most w_e. Ball size: at most $D(D-1)^{k-1}$ vertices at distance $k \geq 1$. ▫

Remarks

1. **Sparse vs dense.** For bounded degree the change is $O(D^{p+1})$ while F grows like m, so the relative change is $O(1/n)$ *and only local terms move*. On the complete DVRP graph A is all edges and B2 reduces to the trivial bound; Lemma A is the right tool there. This contrast is a result to report in Chapter 6.
2. Lemma B is also what makes graph features local (degrees, triangles) meaningful for $p = 1$ (compare Lemma 3).

Theorem C: how far the optimal angles move when the graph changes

Setting

$F = F_G$ and $F' = F_{G'}$ are the objectives before and after the change (old graph embedded with the newisolated, Lemma A0, so both are functions of the same $\theta \in \mathbb{R}^{2p}$). Both are smooth (trigonometric polynomials in θ). θ^* is a local maximiser of F.

Theorem C

Assume that on the ball $\mathcal{B} = \{\theta: \|\theta - \theta^*\| \leq r\}$: 1. (curvature) $\nabla^2 F'(\theta) \preccurlyeq -\mu I$ for some $\mu > 0$; 2. (small gradient change) $\varepsilon := \|\nabla F'(\theta^*)\| = \|\nabla(F' - F)(\theta^*)\|$ satisfies $2\varepsilon/\mu < r$.

Then F' has exactly one local maximiser θ'^* in $\mathcal{B}$, and

$$\| \theta'^* - \theta^* \| \le \frac{\varepsilon}{\mu}.$$

Proof. (The equality in 2 uses $\nabla F(\theta^*) = 0$.)

Existence. Let $\rho = 2\varepsilon/\mu < r$. For θ on the sphere $\| \theta - \theta^* \| = \rho$, Taylor's theorem with assumption 1 gives

$$F'(\theta) \le F'(\theta^*) + \langle \nabla F'(\theta^*), \theta - \theta^* \rangle - \tfrac{\mu}{2}\rho^2 \le F'(\theta^*) + \varepsilon\rho - \tfrac{\mu}{2}\rho^2 = F'(\theta^*).$$

(With ρ slightly larger than $2\varepsilon/\mu$ the inequality is strict; take any such $\rho < r$.) So the maximum of F' on the closed ball of radius ρ is attained in the interior, at a critical point θ'^*.

Uniqueness. F' is strictly concave on $\mathcal{B}$, so it has at most one critical point there.

Distance. Strong concavity gives, for $a, b \in \mathcal{B}$, $\langle \nabla F'(a) - \nabla F'(b), a - b \rangle \le -\mu \| a - b \|^2$. Put $a = \theta^*$, $b = \theta'^*$ ($\nabla F'(b) = 0$): $\mu \| \theta^* - \theta'^* \|^2 \le \langle \nabla F'(\theta^*), \theta'^* - \theta^* \rangle \le \varepsilon \| \theta'^* - \theta^* \|$. Divide. ▫

Bounding ε (connects Theorem C to Lemmas A and B)

ε is the gradient of $F' - F$ at θ^*. The same steps as in Lemma A, applied to $\partial_{\theta_k} F = 2\,\mathrm{Im}\langle \lambda | G_k | \phi \rangle$ (adjoint formula), give a bound of the form $\varepsilon \le c_1 \| \Delta C \| (1 + c_2 \| C \| \sum_l |\gamma_l|)$ with constants depending on p and n (**sketch; full constants not yet derived**). In the sparse regime only the terms in A (Corollary B2) contribute, so $\varepsilon = O(D^{p+1})$. In practice ε and μ are measured directly (below), which is what the thesis needs to compare the regimes.

Numerical check (E3, 2026-11-02)

40 complete Euclidean graphs, $n = 6 \to 7$, $p = 1$, fixed weight scale. Assumption 1 held on the segment in 38 cases. Observed shift $\| \theta'^* - \theta^* \|$: median 0.056 rad. Bound ε/μ: median 0.155. Ratio shift/bound: median 0.31, max 0.68. So the bound held every time and is about 3x conservative; it is much tighter than Lemma A (~14x) because it uses the local curvature.

Remarks

1. Assumption 1 can fail: at a degenerate or flat optimum ($\mu \approx 0$) the optimum can jump. Report how often.
2. Practical meaning: if ε/μ is small, the previous optimum is a good start (temporal warm start); this is the theoretical reason behind RQ3.
3. The cache (C2) uses the same idea with "nearest stored graph" instead of "previous snapshot".

Proposition D and Lemma D0: the first-order (Newton) correction

Setting

As in Theorem C: F (old graph, new vertex isolated) and F' (new graph) are smooth functions of $\theta = (\gamma, \beta) \in \mathbb{R}^{2p}$, θ^* is a local maximiser of F, θ'^* the nearby maximiser of F'. Write $g = \nabla F'(\theta^*)$, $H = \nabla^2 F'(\theta^*)$, $\delta = \theta^* - \theta'^*$ (the "shift").

The **Newton warm start** is

$$\hat{\theta} = \theta^* - H^{-1} g.$$

It is the first-order (implicit-function) correction: it solves the linearised optimality condition $\nabla F'(\theta^*) + H(\theta - \theta^*) = 0$.

Proposition D

Assume on the ball $\mathcal{B}$ of Theorem C (containing θ^* and θ'^*): 1. $\nabla^2 F' \preccurlyeq -\mu I$ (as in Theorem C); 2. the Hessian is M-Lipschitz: $\| \nabla^2 F'(a) - \nabla^2 F'(b) \| \le M \| a - b \|$.

Then

$$\| \hat{\theta} - \theta'^* \| \le \frac{M}{2\mu} \| \theta^* - \theta'^* \|^2 \le \frac{M \varepsilon^2}{2\mu^3}, \qquad \varepsilon = \| \nabla F'(\theta^*) \|.$$

So reusing the old angles has error $\| \delta \|$, while one Newton step has error $O(\| \delta \|^2)$.

Proof. Taylor's theorem with integral remainder for $\nabla F'$ around θ^*:

$$0 = \nabla F'(\theta'^*) = g - H\delta + R, \qquad R = \int_0^1 (\nabla^2 F'(\theta^* - s\delta) - H)(-\delta)\, ds.$$

By assumption 2, $\| R \| \le \int_0^1 Ms \| \delta \|^2 \, ds = \frac{M}{2} \| \delta \|^2$. Hence $g = H\delta - R$ and

$$\hat{\theta} - \theta'^* = \delta - H^{-1} g = \delta - H^{-1}(H\delta - R) = H^{-1} R.$$

By assumption 1 every eigenvalue of H is $\le -\mu$, so $\| H^{-1} \| \le 1/\mu$ and $\| \hat{\theta} - \theta'^* \| \le \frac{M}{2\mu} \| \delta \|^2$. The second inequality is Theorem C ($\| \delta \| \le \varepsilon/\mu$). ▫

A computable (crude) value of M

Centre the cost: $\tilde{C} = C - \frac{C_{\max} + C_{\min}}{2} I$, so $\| \tilde{C} \| = \Delta := \frac{C_{\max} - C_{\min}}{2}$. Replacing C by $\tilde{C}$ changes F by a constant and the circuit by a global phase, so no derivative changes. Each parameter θ_k appears once, as $e^{-i\theta_k G_k}$ with $G_k \in \{\tilde{C}, B\}$, $\| B \| = n$. In the Heisenberg picture, the derivative in θ_k inserts one commutator $i[G_k', \cdot\,]$ with a unitarily conjugated G_k' (same norm), so

$$|\partial_i \partial_j \partial_k F| \le 8 \Delta \| G_i \| \| G_j \| \| G_k \|,$$

and, bounding the operator norm of the third-derivative tensor by its Frobenius norm,

$$M \le 8\Delta(\sum_k \| G_k \|^2)^{3/2} = 8\Delta(p\Delta^2 + pn^2)^{3/2}.$$

This is very loose (E4: the observed constant is ~1000x smaller). It shows that M is finite and polynomial in n, p and the weight scale; the thesis should use the measured constant.

Lemma D0 (a new vertex moves the angles only at second order)

Let the new vertex v have edges with weights $t\,w_{vj}$ (t = a scale, $t = 1$ the real order), and let $F_t(\theta)$ be the QAOA objective of that graph ($F_0 = F$: v isolated). Then for all θ

$$F_t(\theta) = t\frac{W_v}{2} + E_t(\theta), \qquad E_{-t} = E_t, \qquad W_v = \sum_j w_{vj}.$$

Consequently $\nabla_\theta F_t = \nabla_\theta F + O(t^2)$, $\varepsilon = O(t^2)$, and (Theorem C) the shift is $\| \theta'^* - \theta^* \| = O(t^2)$; with Proposition D the Newton error is $O(t^4)$.

Proof. $C_t = C_{\text{old}} + tV$ with $V = \sum_j w_{vj}\frac{1 - Z_v Z_j}{2}$. Conjugation by X_v fixes $|+\rangle^{\otimes n}$, B and C_{old} (no term of C_{old} contains Z_v), and maps $Z_v Z_j \mapsto -Z_v Z_j$, so $X_v V X_v = W_v - V$. Inserting $X_v^2 = I$ everywhere, $F_t(\theta)$ equals the objective with cost $C_{\text{old}} + tW_v - tV$. The constant tW_v is a global phase in the circuit and adds tW_v to the expectation, and $C_{\text{old}} - tV$ is C_{-t}. So $F_t = tW_v + F_{-t}$. Write $F_t = tW_v/2 + E_t$; then $E_t = E_{-t}$. An even smooth function of t has no linear term, and $tW_v/2$ does not depend on θ. ▫

Meaning for the thesis. Adding an order is a “second-order event” for the angles. This is why the observed shifts are small (E3, E4) and why reusing the previous angles (temporal warm start) already works reasonably; it also predicts how the shift grows with the new edge weights (checked numerically: slope 1.97 on a log-log plot, E4). Removing a vertex is the same statement read backwards.

Numerical check (E4, 2026-11-02)

30 complete Euclidean graphs, $n = 6 \to 7$, $p = 1$, reference optima polished to $\| \nabla F \| < 10^{-12}$ with repeated Newton steps. * Lemma D0: shift vs. $t \in \{0.05, 0.1, 0.2\}$, log-log slope median 1.97 (range 1.83 to 2.02). Predicted: 2. * Proposition D: K = err/shift^2 stays bounded, median 2.4, max 3.7 (quadratic behaviour). Bound $M/(2\mu)$ with the crude M: about 1000x larger than K. * Real orders ($t = 1$): shift median 0.052 rad; Newton error median 0.0042 rad; Newton/reuse error ratio median 0.08, max 0.30. Every Newton step was “ok” (Hessian negative definite).

Cost and limits

- Cost: 1 gradient + $2 \cdot 2p$ gradients for the finite-difference Hessian (`perturb.hessian`). On hardware the parameter-shift rule needs $O(p^2)$ circuits: cheap for $p \le 3$, expensive for large p (report this).
- The step needs the curvature assumption. `newton_warm_start` returns the old angles if the Hessian is not negative definite, and caps the step length (status `not_concave` / `capped`).
- Residual learning (C1): the ML model predicts the remaining error $\theta'^* - \hat{\theta}$ instead of θ'^* itself. Proposition D says this residual is $O(\| \delta \|^2)$, so the target is small and well scaled.

Reproducibility

All checks are unit tests in the code repository (`tests/test_stability.py`, `tests/test_locality.py`, `tests/test_angle_shift.py`, `tests/test_perturb.py`), run with fixed random seeds on an exact state-vector simulator.